Prepared for Waverock Holdings, Finance & Operations

Invoice Triage

What the weekly vendor-invoice check does, how to run it, and how it is built.

The 30-file exercise folder, checked Nov 8, 2026

Files read	30
Held for review before payment	14
No issues found	16
Time for a fresh run	about 30 seconds
Amount held for review (USD equivalent)	$116,400.75
All invoices (USD equivalent)	**$146,637.05**

Finance & Operations Associate exercise
November 8, 2026

Summary

Each week accounts payable receives a folder of vendor invoices in mixed formats and has to decide what can go forward for payment and what a person should look at first. Invoice Triage does the reading and the first check, and it can do it on its own every Monday morning.

For every file it produces one row: file, vendor, invoice number, due date, amount, a flag, and the reason for the flag. On the exercise folder it read all 30 files correctly, including the scanned PDF, the phone photo and both emails, and held 14 invoices for review.

What it found

Reason held	Invoices	Detail
Over the $5,000 approval threshold	11	Includes Fathom Research: €4,700 looks under the limit but is about $5,405 once converted.
Duplicate	2	Keystone re-sent invoice KP-2026-0917 by email. Redwood's scan is the same invoice as its PDF, even though the scan prints the number as RW-1O41, with a letter O.
Arithmetic error	1	Summit Office Supply: the lines add up to $1,284.50 but the total due is $1,384.50, a $100 overcharge.
Foreign currency	1	Fathom bills in euros, payable by international wire to a German bank account.
Charge for another company	2	Both Atlas Recruiting placement fees are for roles at other companies (Grasp, PlanIt Geo).

Counts overlap: an invoice can be held for more than one reason. Appendix A lists every file.

How it decides

> **Claude reads; fixed rules decide.** Reading a scan, a photo or a covering email is the part ordinary code does badly, so Claude does it. Whether an invoice is held is decided by plain, tested rules. The same folder gives the same answer every time, and Claude can add a reason to hold an invoice but can never remove one.

What it takes to run

Python 3.10 or newer and Claude Code, signed in; nothing else is installed. A fresh run of 30 files takes about 30 seconds and roughly $0.30 of Claude usage, and unchanged files are never read twice. Nothing is paid, approved or sent anywhere: the tool only prepares the list for a person.

What it does, step by step

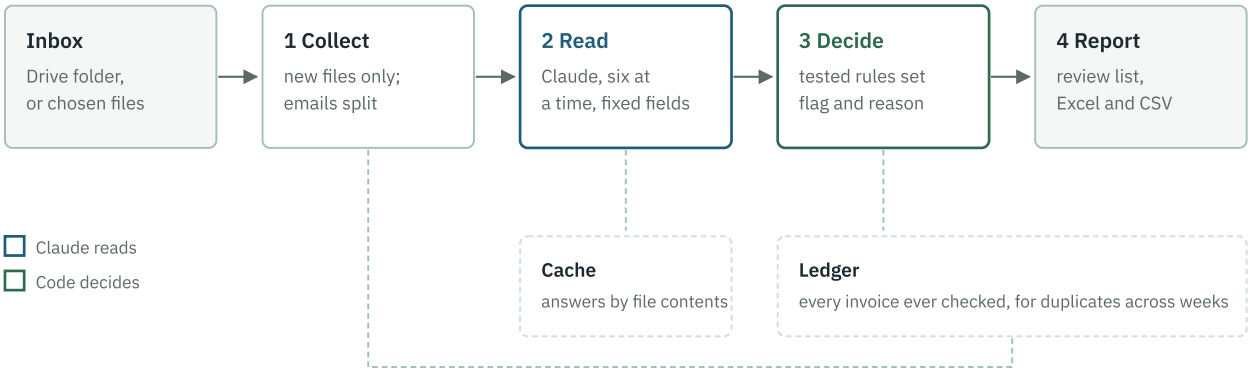


Figure 1. One run. The weekly run is the same path; Collect skips files already in the ledger, and Decide compares each new invoice against it.

1 Collect

Downloads from the shared Drive folder any file it does not already have, or takes the files a clerk chose on the computer. Emails are split into the message and its attachments. A file of an unsupported type or over 25 MB is not dropped: it becomes a held row that says so.

In this folder: 24 digital PDFs, 1 scanned PDF, 3 images and 2 emails.

2 Read

Each document goes to Claude with a fixed list of fields to fill: vendor, invoice number, invoice date exactly as printed, due date if printed, payment terms, currency, each line item, total, who it is billed to, any bank details, what a covering email says, whether the document is a scan, and anything a careful clerk would want checked. Six documents are read at a time.

Claude has no tools and sees nothing but the document. Answers are kept by file contents, so a file that has not changed is never read or paid for twice.

3 Decide

The rules on the next page set the flag and write the reason. Due dates that are not printed are worked out from the payment terms (12 were printed in this folder; the rest came from terms such as Net 30). Duplicates are checked within the folder and against every invoice checked in earlier weeks.

4 Report

The app shows the list with held invoices first. The Excel workbook has an *Invoices* sheet with the seven requested columns and an *Audit* sheet with what sits behind each row: the date as printed, the terms, how the due date was set, currency and conversion, source type and Claude's notes. A CSV copy and the updated ledger are saved with it.

Who does what

Claude	Code
Reads PDFs, scans and photos; copes with any layout	Applies the $5,000 threshold after currency conversion
Reads covering emails for context (“resending as requested”)	Finds duplicates, including look-alike characters such as O and 0
Notices content concerns: changed bank details, charges for another company	Checks every line and the total
Can add a concern to a row	Sets the flag; a concern from Claude can add a reason but never remove one

What gets held for review

Rule	Why it matters	In this folder
Amount over $5,000, after converting to US dollars	Needs approval before payment	11 invoices, including Fathom at €4,700 ≈ $5,405 (rate set in the configuration: 1.15)
Same vendor and invoice number, or same vendor, amount and date	Paying twice is the most common avoidable loss in accounts payable	Keystone email resend; Redwood scan of an invoice already received as a PDF
Lines do not add up to the total, or a line's quantity × price is wrong	Overbilling, often by accident	Summit: $100 over
Not billed in US dollars	Exchange rate and international wire details need confirming	Fathom (EUR)
Billed to someone other than Waverock	Wrong entity, or not our invoice	None
No due date or terms, or a due date before the invoice date	Payment timing cannot be planned	None

Rule	Why it matters	In this folder
A concern Claude raises: changed bank details, a document that says it replaces another, a charge for another company	The common signs of fraud and misdirected invoices	Both Atlas placement fees
A file that could not be read, is too large, or is an email carrying several invoices	Nothing should disappear silently	None

Which copy is the original

When two files are the same invoice, the cleanest copy is kept as the original and every other copy is held: a digital PDF ranks above an email attachment, which ranks above a scan or photo. The original is not held, but its audit row records that a second copy arrived. Invoice numbers are compared after treating look-alike characters as equal, which is how the scan below was matched.

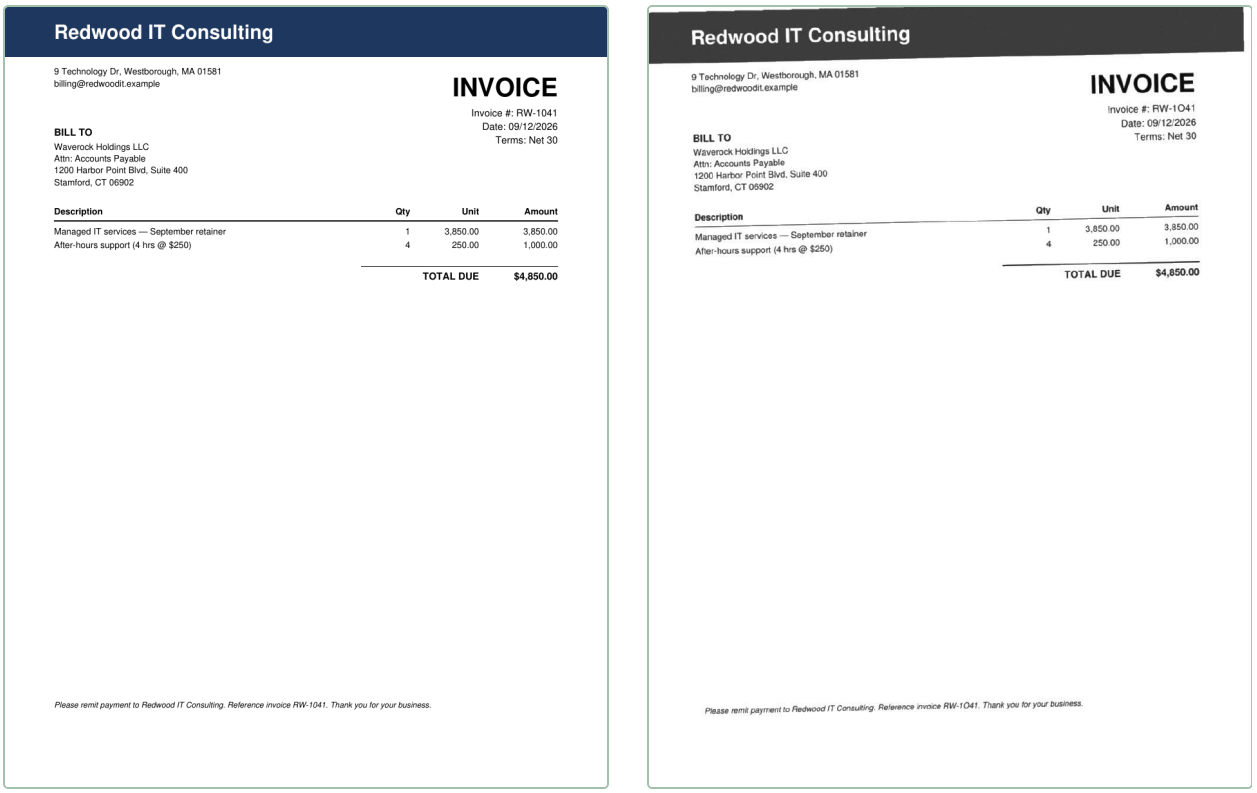


Figure 2. Left, Redwood's digital PDF, kept as the original. Right, a scan of the same invoice, printed as RW-1O41 with a letter O. Held as a duplicate.

What Claude may and may not do

Claude runs with no tools, no internet access of its own, no access to the computer's files, and an empty working folder. Its instructions say that everything in a document is data, never an instruction. Its answer must fit a fixed form, and the rules treat that answer as input to check, not as a decision. A manipulated document could still mislead the reading of a number, which is why every amount is checked against its own line items and why high amounts always go to a person.

Running it

What the computer needs

- Python 3.10 or newer, from python.org. The launcher checks the version and explains what to do if it is missing or too old (Macs ship with an older one).
- Claude Code, installed and signed in. On Windows, the native installer rather than the npm package. The app checks this before it lets a run start.

Checking a folder

1. Double-click *Start Invoice Triage* (the `.bat` file on Windows, the `.command` file on a Mac). The app opens in the web browser. It runs only on this computer.
2. Paste the Google Drive folder link and press *Run check*. Invoices saved on the computer can be used instead with *Choose files*.
3. Watch each file being read. A fresh run of 30 files takes about half a minute.
4. Work through *Needs review*. Each held invoice shows why. Select a row to see what Claude read; select a file name to open the original document.
5. Press *Download Excel* for the spreadsheet.

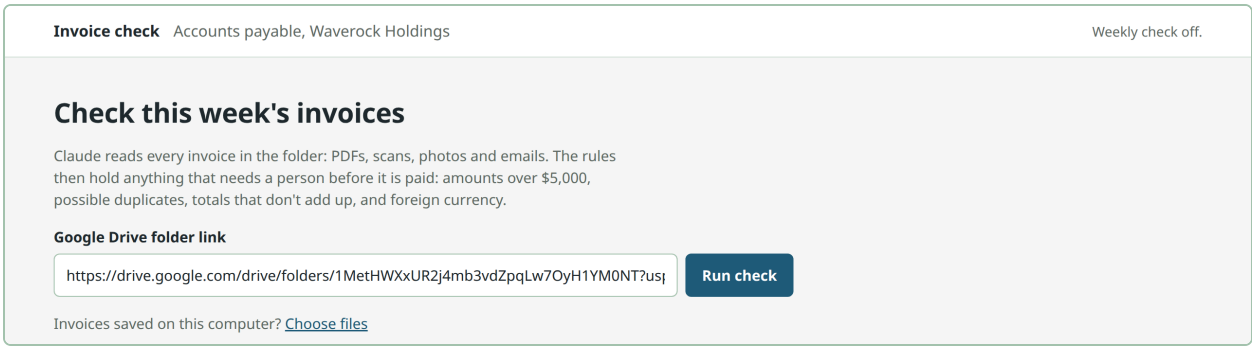
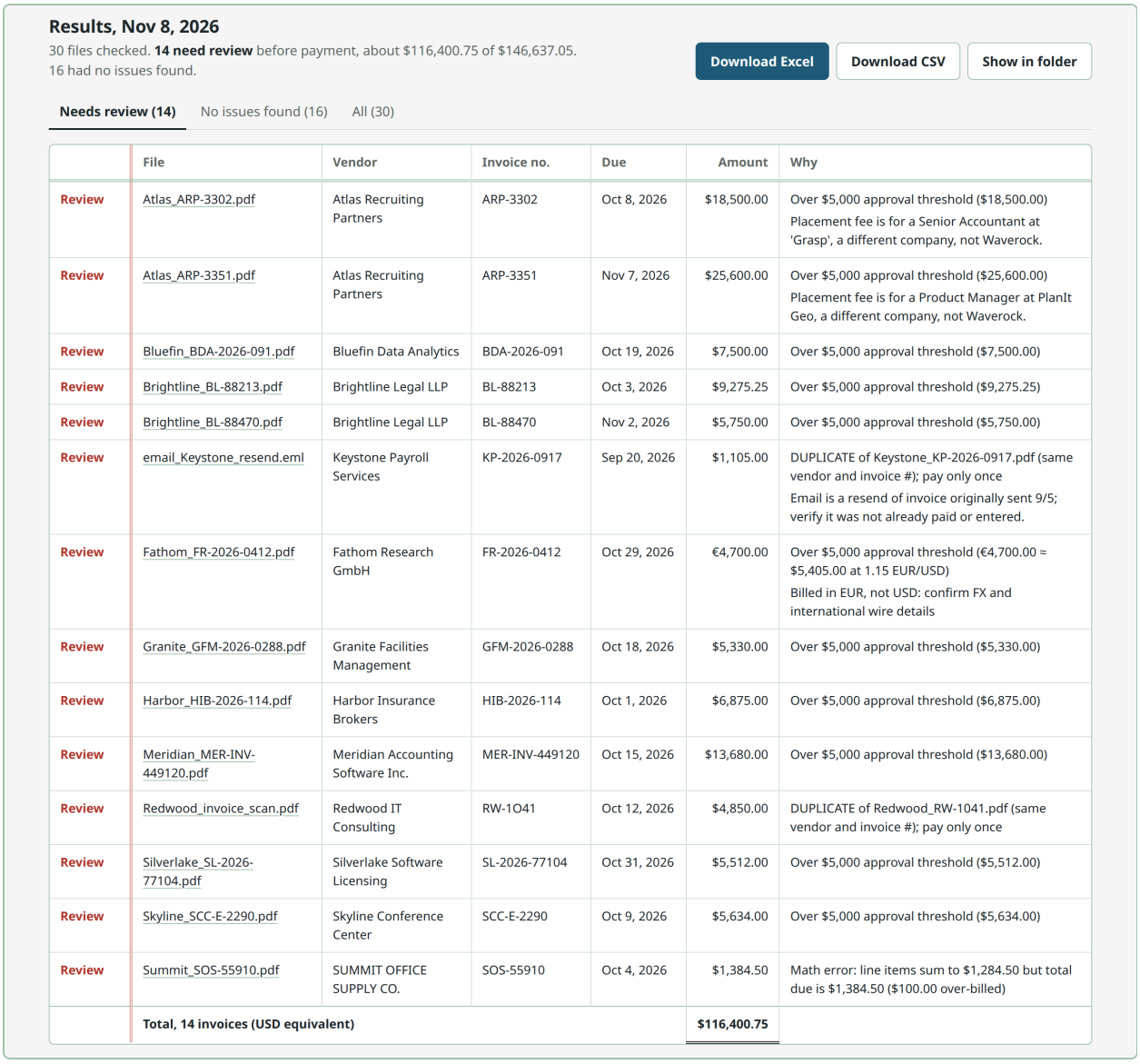


Figure 3. The start screen: one link, one button.



Results, Nov 8, 2026

30 files checked. **14 need review** before payment, about $116,400.75 of $146,637.05. 16 had no issues found.

Download Excel | Download CSV | Show in folder

Needs review (14) | No issues found (16) | All (30)

	File	Vendor	Invoice no.	Due	Amount	Why
Review	Atlas_ARP-3302.pdf	Atlas Recruiting Partners	ARP-3302	Oct 8, 2026	$18,500.00	Over $5,000 approval threshold ($18,500.00) Placement fee is for a Senior Accountant at 'Grasp', a different company, not Waverock.
Review	Atlas_ARP-3351.pdf	Atlas Recruiting Partners	ARP-3351	Nov 7, 2026	$25,600.00	Over $5,000 approval threshold ($25,600.00) Placement fee is for a Product Manager at PlanIt Geo, a different company, not Waverock.
Review	Bluefin_BDA-2026-091.pdf	Bluefin Data Analytics	BDA-2026-091	Oct 19, 2026	$7,500.00	Over $5,000 approval threshold ($7,500.00)
Review	Brightline_BL-88213.pdf	Brightline Legal LLP	BL-88213	Oct 3, 2026	$9,275.25	Over $5,000 approval threshold ($9,275.25)
Review	Brightline_BL-88470.pdf	Brightline Legal LLP	BL-88470	Nov 2, 2026	$5,750.00	Over $5,000 approval threshold ($5,750.00)
Review	email_Keystone_resend.eml	Keystone Payroll Services	KP-2026-0917	Sep 20, 2026	$1,105.00	DUPLICATE of Keystone_KP-2026-0917.pdf (same vendor and invoice #); pay only once Email is a resend of invoice originally sent 9/5; verify it was not already paid or entered.
Review	Fathom_FR-2026-0412.pdf	Fathom Research GmbH	FR-2026-0412	Oct 29, 2026	€4,700.00	Over $5,000 approval threshold (€4,700.00 ≈ $5,405.00 at 1.15 EUR/USD) Billed in EUR, not USD: confirm FX and international wire details
Review	Granite_GFM-2026-0288.pdf	Granite Facilities Management	GFM-2026-0288	Oct 18, 2026	$5,330.00	Over $5,000 approval threshold ($5,330.00)
Review	Harbor_HIB-2026-114.pdf	Harbor Insurance Brokers	HIB-2026-114	Oct 1, 2026	$6,875.00	Over $5,000 approval threshold ($6,875.00)
Review	Meridian_MER-INV-449120.pdf	Meridian Accounting Software Inc.	MER-INV-449120	Oct 15, 2026	$13,680.00	Over $5,000 approval threshold ($13,680.00)
Review	Redwood_invoice_scan.pdf	Redwood IT Consulting	RW-1O41	Oct 12, 2026	$4,850.00	DUPLICATE of Redwood_RW-1041.pdf (same vendor and invoice #); pay only once
Review	Silverlake_SL-2026-77104.pdf	Silverlake Software Licensing	SL-2026-77104	Oct 31, 2026	$5,512.00	Over $5,000 approval threshold ($5,512.00)
Review	Skyline_SCC-E-2290.pdf	Skyline Conference Center	SCC-E-2290	Oct 9, 2026	$5,634.00	Over $5,000 approval threshold ($5,634.00)
Review	Summit_SOS-55910.pdf	SUMMIT OFFICE SUPPLY CO.	SOS-55910	Oct 4, 2026	$1,384.50	Math error: line items sum to $1,284.50 but total due is $1,384.50 ($100.00 over-billed)
	Total, 14 invoices (USD equivalent)				**$116,400.75**	

Figure 4. Results for the exercise folder, held invoices first. The red double rule is the margin of a ledger pad: the status sits in the margin, the reason on the right, and the column of amounts closes with a double-ruled total.

The two words in the margin are deliberate. *Review* means a person must look before payment. *No issues found* means none of the rules fired; it is not an approval, and the tool never says an invoice is cleared to pay.

Every Monday, without anyone

Turning on *Check the Drive folder every Monday at 07:00* registers a job with the computer's own scheduler. Nothing extra is installed and nothing has to stay open.

System	Scheduler used	If the computer was off at 07:00
Windows	Task Scheduler, without a console window	Runs as soon as possible afterwards
macOS	launchd (a user LaunchAgent)	Runs when the Mac wakes
Linux	systemd user timer	Runs at the next start (Persistent)

What the Monday run does

- Fetches the Drive folder and reads only files it has not seen before, judged by name and contents.
- Compares every new invoice with the ledger of everything ever checked, so a vendor's resend next month is still caught, even under a new file name.
- Saves that week's spreadsheet as *Weekly results/Invoices <date>.xlsx* and shows the result in the app.

When something goes wrong

- A run that cannot finish writes *Weekly results/LAST_RUN_FAILED.txt*, saying what happened and how to fix it in plain words. The app shows the same message. The note is removed after the next good run.
- If no file at all can be read, which usually means Claude Code is signed out or the connection is down, the run counts as failed rather than producing an empty list.
- Only one run happens at a time. A manual check and the Monday job cannot overwrite each other, and a lock left behind by a closed window is cleared automatically.

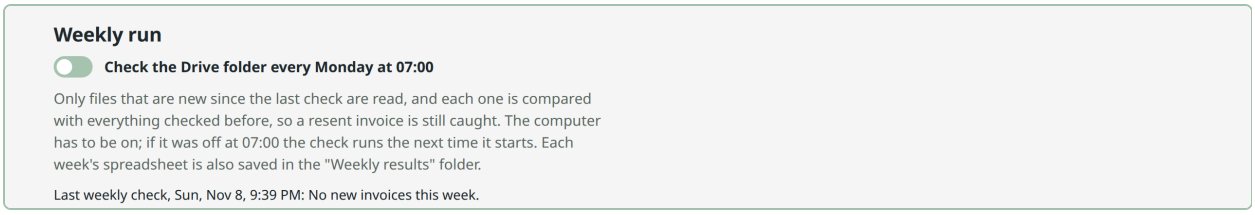


Figure 5. The weekly switch and the status of the last Monday run.

Taking it off a personal computer

The scheduler approach depends on one computer being on. The same code runs unchanged on a server or in a scheduled GitHub Actions job when an Anthropic API key is provided; Claude is then called in API mode, with no sign-in. For production I would make three changes: read the folder with a Google service account instead of a public link, write results straight into the team's shared Sheet, and post the weekly summary to the AP channel.

How it is built

Design decisions

The AI reads, the code decides. Every flag comes from rules in one file (`rules.py`) with unit tests. Claude's answer is input to those rules, never the decision.

A locked-down model call. Each document is sent to the local Claude Code command line with tools, plug-ins and personal settings switched off, in an empty folder, and the answer must match a fixed form. The cache is keyed by file contents and by the exact instructions, so editing the instructions can never return stale answers.

Nothing disappears. A file that cannot be read becomes a held row that says why. Results, cache and ledger are written so that a crash never leaves a half-written file. One run at a time.

No installation. Python's standard library only. The Excel workbook is written directly, without spreadsheet libraries, and cells that start like a formula are neutralised so a vendor's text can never execute in Excel.

Components

Part	Responsibility
`pipeline.py`	One run from start to finish; run lock; ledger
`drive.py`, `ingest.py`	Fetch the folder; turn files and emails into what Claude reads; type and size limits
`claude_cli.py`, `extract.py`	Find and call Claude Code safely on each system; the fixed form; the cache
`rules.py`	Every flag and every reason
`report.py`	Excel, CSV and summary
`scheduler.py`, `weekly.py`	The Monday job on Windows, macOS and Linux; failure notes
`server.py`, `web/`	The local app in the browser

The local app is private to the computer

It listens only on the computer itself and refuses requests addressed to any other name. Every request must carry a secret created each time the app starts, so another website cannot drive it. The page will not load outside scripts and shows document text as text, never as page code. Uploaded files keep only their name, must be PDF, image or email, and are capped at 25 MB. Only files that belong to a run can be opened from it.

How it was checked

Check	Result
Three fresh runs compared field by field with a hand-checked answer key (invoice number, due date, amount and flag for all 30 files)	0 mismatches in each run
Automated tests	54 passing: rules, duplicates, arithmetic, currency, due dates, email handling, scheduler files for all three systems, app security, and a full run end to end
Linux scheduler files	Validated with systemd's own checker
Independent review of the plan, before building	14 changes adopted, among them scan-versus-PDF ranking, the run lock and Windows scheduler settings
Independent review of the finished build	12 findings; 10 fixed, each with a test, among them a stale lock after a closed window and a sign-in check that only checked installation

Limits and next steps

Known limits

- Tested on Linux. The Windows and macOS paths are written and their scheduler files tested, but they have not yet run on a Windows PC or a Mac.
- The euro rate is a fixed number in the configuration.
- All-numeric dates are read as month/day/year; the printed date is kept in the Audit sheet.
- One row per file; an email carrying several invoices is held for a person to split.
- A corrected file re-uploaded to Drive under the same name is not fetched again.
- The Monday run needs the computer on, or catches up at the next start.

Next steps

- Run on a server with an API key; read Drive with a service account; write to the shared Sheet.
- A vendor master, to hold new vendors and any change of bank details.
- A daily exchange-rate feed.
- For digital PDFs, check Claude's total against the PDF's own text as a second, independent reading.
- A weekly summary posted to the AP channel.

Appendix A. Every file in the exercise folder

Status	File, and why it is held	Vendor	Invoice no.	Due	Amount
Review	**Atlas_ARP-3302.pdf** Over $5,000 approval threshold ($18,500.00) Placement fee is for a Senior Accountant at 'Grasp', a different company, not Waverock.	Atlas Recruiting Partners	ARP-3302	Oct 8, 2026	$18,500.00
Review	**Atlas_ARP-3351.pdf** Over $5,000 approval threshold ($25,600.00) Placement fee is for a Product Manager at PlanIt Geo, a different company, not Waverock.	Atlas Recruiting Partners	ARP-3351	Nov 7, 2026	$25,600.00
Review	**Bluefin_BDA-2026-091.pdf** Over $5,000 approval threshold ($7,500.00)	Bluefin Data Analytics	BDA-2026-091	Oct 19, 2026	$7,500.00
Review	**Brightline_BL-88213.pdf** Over $5,000 approval threshold ($9,275.25)	Brightline Legal LLP	BL-88213	Oct 3, 2026	$9,275.25
Review	**Brightline_BL-88470.pdf** Over $5,000 approval threshold ($5,750.00)	Brightline Legal LLP	BL-88470	Nov 2, 2026	$5,750.00
No issues	**Copperline_CP-10233.pdf**	Copperline Print & Design	CP-10233	Oct 7, 2026	$788.00
Review	**email_Keystone_resend.eml** DUPLICATE of Keystone_KP-2026-0917.pdf (same vendor and invoice #); pay only once Email is a resend of invoice originally sent 9/5; verify it was not already paid or entered.	Keystone Payroll Services	KP-2026-0917	Sep 20, 2026	$1,105.00
No issues	**email_Lumen_invoice.eml**	Lumen Marketing Studio	LMS-0781	Oct 11, 2026	$4,470.00
No issues	**Evergreen_EBA-55021.pdf**	Evergreen Benefits Administration	EBA-55021	Oct 24, 2026	$1,529.00
Review	**Fathom_FR-2026-0412.pdf** Over $5,000 approval threshold (€4,700.00 ≈ $5,405.00 at 1.15 EUR/USD) Billed in EUR, not USD: confirm FX and international wire details	Fathom Research GmbH	FR-2026-0412	Oct 29, 2026	€4,700.00
Review	**Granite_GFM-2026-0288.pdf** Over $5,000 approval threshold ($5,330.00)	Granite Facilities Management	GFM-2026-0288	Oct 18, 2026	$5,330.00
No issues	**Granite_GFM-2026-0301.pdf**	Granite Facilities Management	GFM-2026-0301	Nov 8, 2026	$2,150.00
Review	**Harbor_HIB-2026-114.pdf** Over $5,000 approval threshold ($6,875.00)	Harbor Insurance Brokers	HIB-2026-114	Oct 1, 2026	$6,875.00
No issues	**IMG_2026013.jpg**	CLEARWATER COFFEE SERVICE	CW-4410	Sep 17, 2026	$228.00

Status	File, and why it is held	Vendor	Invoice no.	Due	Amount
No issues	**Ironwood_ISS-30917.pdf**	IRONWOOD SECURITY SYSTEMS	ISS-30917	Oct 26, 2026	$1,125.00
No issues	**Keystone_KP-2026-0917.pdf**	Keystone Payroll Services	KP-2026-0917	Sep 20, 2026	$1,105.00
No issues	**Keystone_KP-2026-1044.pdf**	Keystone Payroll Services	KP-2026-1044	Oct 21, 2026	$830.00
Review	**Meridian_MER-INV-449120.pdf** Over $5,000 approval threshold ($13,680.00)	Meridian Accounting Software Inc.	MER-INV-449120	Oct 15, 2026	$13,680.00
No issues	**Northwind_NW-2026-10447.pdf**	Northwind Cloud Hosting	NW-2026-10447	Oct 2, 2026	$3,826.40
No issues	**Northwind_NW-2026-10992.pdf**	Northwind Cloud Hosting	NW-2026-10992	Nov 1, 2026	$3,742.50
No issues	**Oakridge_OEC-0917.pdf**	Oakridge Executive Coaching	OEC-0917	Oct 30, 2026	$1,350.00
No issues	**Orion_OT-7731902.pdf**	ORION TELECOM	OT-7731902	Oct 10, 2026	$998.00
Review	**Redwood_invoice_scan.pdf** DUPLICATE of Redwood_RW-1041.pdf (same vendor and invoice #); pay only once	Redwood IT Consulting	RW-1O41	Oct 12, 2026	$4,850.00
No issues	**Redwood_RW-1041.pdf**	Redwood IT Consulting	RW-1041	Oct 12, 2026	$4,850.00
No issues	**scan_006.png**	PINECREST JANITORIAL SERVICES	PJ-0926	Sep 9, 2026	$980.00
No issues	**scan_027.png**	PINECREST JANITORIAL SERVICES	PJ-1026	Oct 6, 2026	$925.00
Review	**Silverlake_SL-2026-77104.pdf** Over $5,000 approval threshold ($5,512.00)	Silverlake Software Licensing	SL-2026-77104	Oct 31, 2026	$5,512.00
Review	**Skyline_SCC-E-2290.pdf** Over $5,000 approval threshold ($5,634.00)	Skyline Conference Center	SCC-E-2290	Oct 9, 2026	$5,634.00
Review	**Summit_SOS-55910.pdf** Math error: line items sum to $1,284.50 but total due is $1,384.50 ($100.00 over-billed)	SUMMIT OFFICE SUPPLY CO.	SOS-55910	Oct 4, 2026	$1,384.50
No issues	**Vantage_VTM-20260915-07.pdf**	Vantage Travel Management	VTM-20260915-07	Sep 30, 2026	$1,339.40

Appendix B. Reference

Command line

Everything the app does can also be run from a terminal in the project folder.

```
python3 -m invoice_triage                                  open the app
python3 -m invoice_triage run --drive <folder link>      check a Drive folder
python3 -m invoice_triage run --folder <path>            check a local folder
python3 -m invoice_triage run --drive <link> --only-new  weekly behaviour, by hand
python3 -m invoice_triage weekly                          what the Monday job runs
python3 -m unittest discover -s tests -t .                run the tests
```

Settings

`config.json` holds the approval threshold, exchange rates, the company names accepted as the bill-to, the Claude model and how many documents are read at once.

Where things are saved

Folder	Contents
`runs/<date and time>/`	Each run: the Excel workbook, CSV, summary and the full results
`Weekly results/`	Each Monday's workbook, and a failure note if a run could not finish
`state/`	The ledger, the cache, downloaded Drive files and the app's settings